

RNT2

General Physics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 0.6 kg rock is dropped from rest off a footbridge built in 1962. It hits the water 3.0 s later. Ignoring air resistance and using $g = 9.8 \text{ m/s}^2$, how high is the bridge above the water?

- A. 14.7 m
- B. 88.2 m
- C. 44.1 m
- D. 29.4 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A warehouse motor rated for 20,000 cycles lifts a 50 kg crate straight up 12 m in 8 s at constant speed. The warehouse runs three shifts. What average power does the motor deliver? Use $g = 9.8 \text{ m/s}^2$.

- A. 735 W
- B. 5,880 W
- C. 47,040 W
- D. 75 W

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A cook heats 2.0 kg of water from 20 °C to 80 °C in a stainless pot on a 1,500 W burner. Water's specific heat is 4,186 J/(kg·°C). How much heat does the water absorb?

- A. About 140 J
- B. About 670,000 J
- C. About 502,000 J
- D. 8,372 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A 0.50 kg block of aluminum ($c = 900 \text{ J}/(\text{kg}\cdot^\circ\text{C})$) starts at 25°C and absorbs 9,000 J. What is its temperature change?

- A. 10°C
- B. 20°C
- C. 5°C
- D. 45°C

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A gas in a steel cylinder with a 10 cm diameter absorbs 500 J of heat and does 200 J of work pushing a piston outward. The lab keeps the cylinder in a locked cabinet. What is the change in the gas's internal energy?

- A. +300 J
- B. +500 J
- C. +700 J
- D. 0 J, because energy is always conserved

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A gas's internal energy decreases by 150 J while 400 J of work is done on the gas. What is the heat exchange?

- A. The gas absorbs 250 J of heat.
- B. The gas absorbs 550 J of heat.
- C. The gas releases 150 J of heat.
- D. The gas releases 550 J of heat.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A hobbyist connects a $4\ \Omega$ resistor and an $8\ \Omega$ resistor in series across a 24 V battery using copper wires 2 m long. She bought the kit at a weekend fair. What current flows from the battery?

- A. 2 A
- B. 3 A
- C. 9 A
- D. 288 A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Two $6\ \Omega$ resistors are connected in parallel across a 12 V battery. What total current flows from the battery?

- A. 2 A
- B. 1 A
- C. 72 A
- D. 4 A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Two small charged spheres on nylon insulating stands are 2 m apart and repel with force F . A technician with a new multimeter moves them to 6 m apart without changing the charges. What is the new force?

- A. $F/3$
- B. $F/9$
- C. $9F$
- D. F , because the charges did not change

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

The force between two fixed charges becomes 4 times larger. The charges did not change. What happened to the distance between them?

- A.** It did not change; only a change in charge can do that.
- B.** It was cut in half.
- C.** It was cut to one-fourth.
- D.** It doubled.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A 200 W red lamp placed 3 m from a clean zinc plate frees no electrons, even after 10 minutes. The lab bought the lamp from a theater supplier. What change is most likely to cause electrons to be emitted?

- A.** Nothing; metals never emit electrons under light.
- B.** Leave the red lamp on longer.
- C.** Use ultraviolet light, which has a higher frequency.
- D.** Use a brighter red lamp.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A hospital stores 200 g of an isotope in a lead container. The isotope has a half-life of 8 days. The container weighs 30 kg. How much of the isotope remains after 24 days?

- A.** About 66.7 g
- B.** 50 g
- C.** 0 g, since three half-lives use it up
- D.** 25 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A ~~0.6 kg rock~~ is dropped from rest off a footbridge ~~built in 1962~~. It hits the water 3.0 s later. Ignoring air resistance and using $g = 9.8 \text{ m/s}^2$, how high is the bridge above the water?

Target word: fall distance **Decides:** $d = \frac{1}{2} \times 9.8 \times 3.0^2$.

A	Adjacent category	$\frac{1}{2} \times 9.8 \times 3 = 14.7$ is the average speed in m/s; it still has to be multiplied by the time.
B	Dropped element	Drops the $\frac{1}{2}$.
C	Correct	$\frac{1}{2} \times 9.8 \times 9 = 44.1 \text{ m}$.
D	Confusion set	$9.8 \times 3 = 29.4$ is the final speed in m/s, not a distance.

2. Answer A Concept 2

A warehouse motor ~~rated for 20,000 cycles~~ lifts a 50 kg crate straight up 12 m in 8 s at constant speed. ~~The warehouse runs three shifts.~~ What average power does the motor deliver? Use $g = 9.8 \text{ m/s}^2$.

Target word: power **Decides:** $W = mgh = 50 \times 9.8 \times 12 = 5,880 \text{ J}$; $P = W / t$.

A	Correct	$5,880 \text{ J} / 8 \text{ s} = 735 \text{ W}$.
B	Dropped element	That is the work in joules; dividing by time is dropped.
C	Inverted operation	Multiplies the work by the time instead of dividing.
D	Confusion set	Uses mass (50 kg) as the force instead of weight (490 N).

3. Answer C Concept 3

A cook heats 2.0 kg of water from 20 °C to 80 °C ~~in a stainless pot on a 1,500-W burner~~. Water's specific heat is 4,186 J/(kg·°C). How much heat does the water absorb?

Target word: heat absorbed **Decides:** $\Delta T = 60 \text{ °C}$; $Q = 2.0 \times 4,186 \times 60$.

A	Inverted operation	Divides by ΔT instead of multiplying.
B	Wrong stem number	Uses the final temperature, 80 °C, instead of the change.
C	Correct	$2.0 \times 4,186 \times 60 = 502,320 \text{ J}$.
D	Dropped element	Drops ΔT : $2.0 \times 4,186$.

4. Answer B Concept 3 · transfer

A 0.50 kg block of aluminum ($c = 900 \text{ J/(kg·°C)}$) starts at 25 °C and absorbs 9,000 J. What is its temperature change?

Target word: solve for ΔT **Decides:** $\Delta T = Q / (mc) = 9,000 / (0.50 \times 900)$.

A	Dropped element	Drops the mass: $9,000 / 900$.
B	Correct	$9,000 / 450 = 20 \text{ °C}$.
C	Inverted operation	Multiplies by the mass instead of dividing: $9,000 \times 0.50 / 900$.
D	Confusion set	That is the final temperature, not the change.

5. Answer A Concept 4

A gas in a steel cylinder with a 10 cm diameter absorbs 500 J of heat and does 200 J of work pushing a piston outward. ~~The lab keeps the cylinder in a locked cabinet.~~ What is the change in the gas's internal energy?

Target word: ΔU **Decides:** $\Delta U = Q - W_{\text{by}} = 500 - 200$.

A	Correct	Heat in raises U; work done by the gas lowers it.
B	Dropped element	Drops the work term.
C	Inverted operation	Adds the work done by the gas.
D	Absolute language	Conservation is exactly why U changes by $Q - W$.

6. Answer D Concept 4 • transfer

A gas's internal energy decreases by 150 J while 400 J of work is done on the gas. What is the heat exchange?

Target word: solve for Q **Decides:** $\Delta U = Q + W_{\text{on}}$: $-150 = Q + 400$.

A	Inverted operation	Treats $400 - 150$ as heat in, getting the sign of the work wrong.
B	Confusion set	Right size, wrong direction.
C	Dropped element	Drops the work term.
D	Correct	$Q = -550$ J.

7. Answer A Concept 5

A hobbyist connects a 4 Ω resistor and an 8 Ω resistor in series across a 24 V battery ~~using copper wires 2 m long. She bought the kit at a weekend fair.~~ What current flows from the battery?

Target word: series current **Decides:** $R = 4 + 8 = 12 \Omega$.

A	Correct	$24 \text{ V} / 12 \Omega = 2 \text{ A}$.
B	Dropped element	Uses only the 8 Ω resistor.
C	Confusion set	Uses the parallel equivalent, $8/3 \Omega$.
D	Inverted operation	Multiplies $V \times R$ instead of dividing.

8. Answer D Concept 5 • transfer

Two 6 Ω resistors are connected in parallel across a 12 V battery. What total current flows from the battery?

Target word: parallel current **Decides:** Two equal 6 Ω in parallel give 3 Ω .

A	Dropped element	Current through one branch only.
B	Confusion set	Uses the series total, 12 Ω .
C	Inverted operation	Multiplies $V \times R$.
D	Correct	$12 \text{ V} / 3 \Omega = 4 \text{ A}$.

9. Answer B Concept 6

Two small charged spheres ~~on nylon insulating stands~~ are 2 m apart and repel with force F. A technician ~~with a new multimeter~~ moves them to 6 m apart without changing the charges. What is the new force?

Target word: force after distance change **Decides:** Distance tripled, so F is divided by $3^2 = 9$.

A	Dropped element	Drops the square.
B	Correct	Inverse square: $(1/3)^2 = 1/9$.
C	Inverted operation	Force falls, not rises, with distance.
D	Absolute language	Force depends on distance as well as charge.

10. Answer B Concept 6 · transfer

The force between two fixed charges becomes 4 times larger. The charges did not change. What happened to the distance between them?

Target word: solve for the distance change **Decides:** $4 = 1/k^2$, so $k = 1/2$.

A	Adjacent category	Distance changes force through the inverse square.
B	Correct	Halving r multiplies F by $2^2 = 4$.
C	Dropped element	Drops the square; that would make F 16 times larger.
D	Inverted operation	Doubling r makes F four times smaller.

11. Answer C Concept 7

A 200 W red lamp placed 3 m from a clean zinc plate frees no electrons, ~~even after 10 minutes. The lab bought the lamp from a theater supplier.~~ What change is most likely to cause electrons to be emitted?

Target word: what triggers emission **Decides:** Emission depends on frequency, not intensity.

A	Absolute language	They do above the threshold frequency.
B	Camouflage	Photon energy, not exposure time, decides.
C	Correct	Each photon then carries enough energy (hf) to free an electron.
D	Adjacent category	More red photons, but each still has too little energy.

12. Answer D Concept 8

A hospital stores 200 g of an isotope ~~in a lead container.~~ The isotope has a half-life of 8 days. ~~The container weighs 30 kg.~~ How much of the isotope remains after 24 days?

Target word: amount remaining **Decides:** $24 / 8 = 3$ half-lives.

A	Adjacent category	Linear: $200 / 3$.
B	Dropped element	Counts only two half-lives.
C	Absolute language	Each half-life halves the amount; it never reaches zero by subtraction.
D	Correct	$200 \rightarrow 100 \rightarrow 50 \rightarrow 25$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	11	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Free-fall kinematics	An object dropped from rest falls $d = \frac{1}{2}gt^2$ and reaches $v = gt$, with $g \approx 9.8 \text{ m/s}^2$, independent of mass.	Dropping the $\frac{1}{2}$, or giving speed when distance is asked.	“Dropped” and a fall time.
2	Work and power	Lifting work is $W = mgh$ (weight $\times$ height); power is $P = W / t$.	Using mass as the force, or reporting work as power.	A lift height and a time.
3	Specific heat	$Q = mc\Delta T$, where ΔT is the change in temperature, not the final temperature.	Using the final temperature for ΔT , or leaving out the mass.	Heating or cooling a mass by a stated temperature range.
4	First law of thermodynamics	$\Delta U = Q - W$, where Q is heat added to the system and W is work done by the system.	Adding the work the gas does, or ignoring work entirely.	A gas that absorbs heat and does work, or has work done on it.
5	Series and parallel resistors	Series: $R = R_1 + R_2$; parallel: $1/R = 1/R_1 + 1/R_2$; then $I = V / R$.	Using the parallel rule for series, or the reverse.	“In series” or “in parallel” with a battery voltage.
6	Coulomb’s law inverse-square scaling	$F \propto q_1q_2 / r^2$; multiplying r by k divides F by k^2 .	Scaling force by k instead of k^2 .	The distance between charges changes by a factor.
7	Photoelectric effect	Emission depends on light frequency (photon energy $E = hf$), not intensity; below the threshold frequency, no electrons come out however bright the light is.	Thinking brighter or longer exposure will eventually free electrons.	A metal that emits no electrons under a bright light.
8	Radioactive half-life	Each half-life cuts the remaining amount in half: $N = N_0(1/2)^{(t / T)}$.	Subtracting a fixed amount per half-life (linear decay).	A half-life and an elapsed time.

Course scope (public description): This course provides a broad overview of the principles of mechanics, thermodynamics, wave motion, modern physics, and electricity and magnetism and invites students to apply them by solving problems, performing labs, and reflecting on concepts and ideas.